

What is Physiotherapy?

Physiotherapy is a form of treatment carried out through the medium of physical forces such as heat, electricity, mechanical pressure, and mechanical forces. Thus, in physiotherapy, heat, electrical current, water, soft tissue manipulation, and exercises with or without resistance are utilized. Physiotherapy is a health profession whose primary purpose is the promotion of optimal human health by the application of scientific principles to prevent, assess, correct, or alleviate acute or prolonged movement dysfunction.

These are several methods of the physiotherapy:

- 1- Infrared rays (IR) & Ultraviolet (UV) light.
- 2- Short-wave & Microwave diathermy
- 3- Electrical stimulation therapy
- 4- Ultrasound therapy.

1- Infrared (IR) & Ultraviolet (UV) light Therapy

Infrared light is electromagnetic radiation with longer wavelengths than those of visible light.

Much of energy from the sun arrives on earth in the form of infrared radiation. Infrared radiation is subdivided into three regions, A, B and C. They are approximately distinguished by their absorption characteristics. When infrared radiation strikes tissue, it causes molecules to vibrate, producing heat and causing the temperature to rise.

Infrared light used for what?

- A and B are utilized therapeutically, A is near or short wavelength radiation (760-1400) nm, and B is far or long wavelength (1400-3000) nm.



Short infrared and red visible radiation have penetration depths about (2-3) mm, infrared radiation is strongly absorbed near the skin surface, and little is reflected from the surface if it is applied perpendicular to the skin. The therapeutic effects of heat

Infrared [radiation](#) can promote local blood circulation and reduce muscle tension. Examples of traditional medical applications of infrared [radiation](#) include the relief of Back and Neck pain muscle pain, arthritis and tension,

Risk of infrared light:

- The main danger of infrared treatment is a burn , burns are always a potential risk if heat is too intense.
- There is very low level risk of eye damage.

Ultraviolet light therapy (UV):

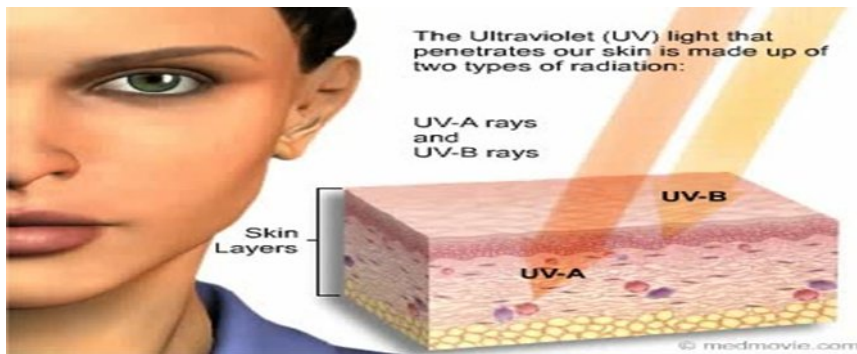
Ultraviolet light therapy is also known as phototherapy , it is a kind of therapy that involves exposing the skin to Ultraviolet light for a prescribed duration of time.

The use of ultraviolet light therapy has proven to be very useful for treating various kinds of skin problems such as psoriasis, eczema , and other skin conditions.

Ultraviolet light is subdivided into three regions, A, B and C. Phototherapy can be applied directly on the skin by the therapist.

UVB rays have a short wavelength that reaches the outer layer of your skin (the epidermis)

UVA rays have a longer wavelength that can penetrate the middle layer of your skin (the dermis)



UV light used to treat a mild deficiency of vitamin D. It may also involve making use of medications or chemical agents that can be applied on the skin to increase the effect of the applied UV light that is used for treatment.

Benefits of Ultraviolet light therapy:

- Helps improve blood circulation in the affected area.
- Stimulates production of collagen required for repairing damaged tissues.

The effect of Ultraviolet light (Risk):

Strong doses of ultraviolet to the eye can causes conjunctivitis.

2- Short –wave & Microwave diathermy

Diathermy is the application of high-frequency electromagnetic energy that is primarily used to generate heat in body tissues. Heat is produced by resistance of the tissue to the passage of the energy. Diathermy may also be used to produce non thermal effects.

Short-wave is the first methods which transferring electromagnetic energy into the body, the part of body treated is placed between two metal plate like electrodes energized by the high frequency voltage. Shortwave therapy equipment commonly used the frequency of 27.12 MHz This corresponds to a wavelength of approximately 11 meters.

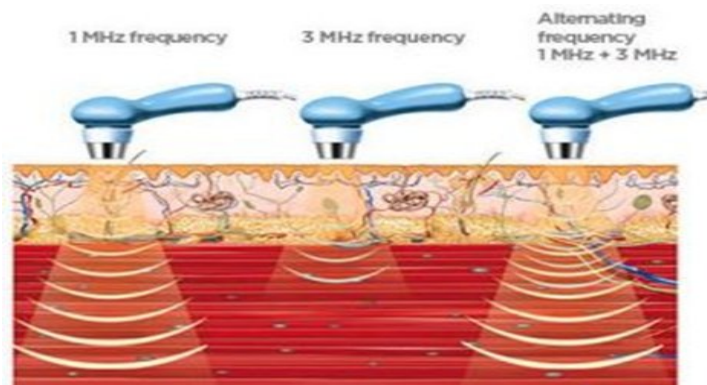


Microwave diathermy, another form of electromagnetic energy, is usually easier to apply than short-wave diathermy. The microwaves are obtained by heating special tube called Magnetron. Microwave emitted from the applicator antenna, which can be placed several inches from the region to be treated. Microwaves can penetrate deep into the tissue causing heating. The frequency used is 900 MHz, it causes more uniform heating around bony regions.

When the electromagnetic energy of the microwave radiation is absorbed in the tissue, the amount of heat produced inside the tissue depends upon the fluid contents. The effective depth of penetration of MWD is 4.3 cm. As Microwaves are more absorbed by the water so those tissues which have good blood supply like muscle are heated mostly while those tissues which have low fluid contents like fat produce less heat.

3- Ultrasonic Physiotherapy

Ultrasonic waves are a form of mechanical energy or sound energy due to mechanical vibrations at very high frequency. Normal human can hear sounds from 20 to 20 000 Hz while the frequencies used in therapeutic medical purposes are beyond human sound range and are usually around 1MHz or 3 MHz : 1 MHz = 1 million cycles per sec.



How does it work?

The ultrasound transducer is the part of the device that is in contact with the skin. By using two types of transducers with different frequencies, 1MHz, and 3MHz according to the area of treatment. This transducer is moved continuously over the skin for approximately (3-5) minutes. The Power should be greater than 1.2 W/cm^2 for fatty tissue such as the thighs, and less than 1.2 W/cm^2 for low-fat tissue such as the fingers. Continuous Sound wave energy will cause oscillations in the particles in that material. Assuming this material to be human tissue, any increase in molecular vibration will result in heat generation, which can produce thermal changes in the tissue. This heat therapy is used in physiotherapy equipment called ultrasonic machines.

4- Electrical stimulation therapy

Electrical stimulation uses an electrical current to cause a single or a group of muscles to contract. By placing electrodes on the skin in various locations to make it ready for functional use.

Electrical stimulation therapy has three types; general, muscular, and transcutaneous electrical nerve stimulation (TENS). General electric therapy is used for alleviating pain and healing wounds. Muscular electric stimulation aims to strengthen muscles by reducing muscle spasms. TENS is used to treat chronic pain.

What is TENS Machine?

There are multiple leads based on several channels and electrodes to make it ready for functional use. This treatment is used for pain relief and involves a high-frequency signal applied to the painful area. Basically, it stimulates the nerve which cuts off this pain message to the spinal cord or brain. TENS is used for osteoarthritis, joint and muscle pain, back pain, migraines, and neuralgic pain



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